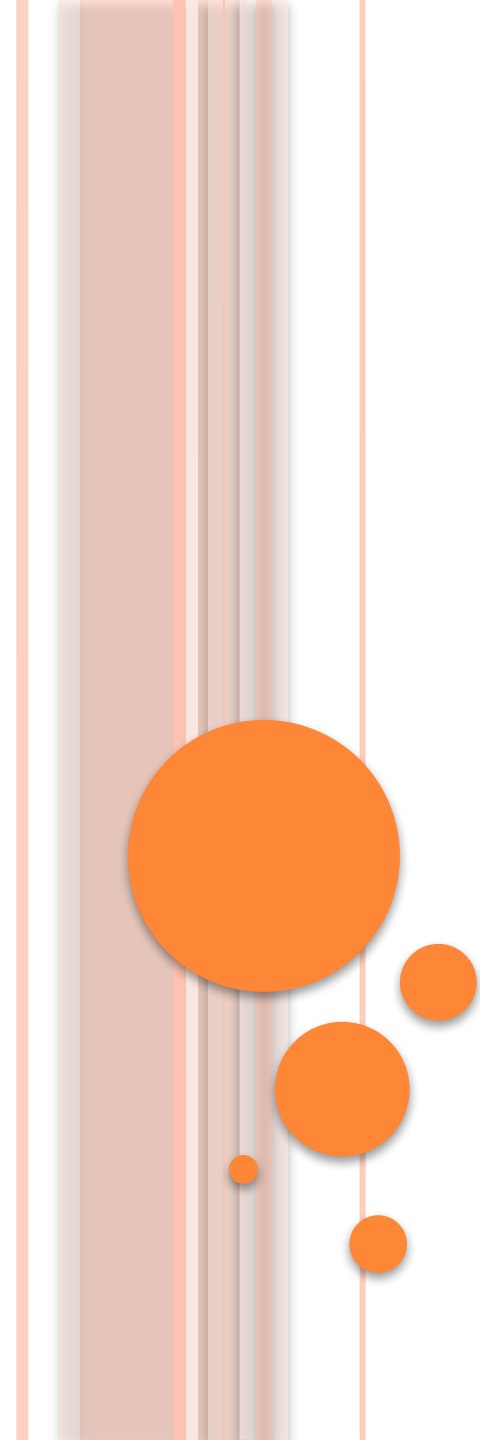




CHAPTER 2

HEALTH COMMODITY SECURITY

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OVERVIEW OF THE LESSON

- Health commodity security
- Strategic pathway to health commodity security
- Steps involved in health commodity security
- The purpose of logistics system
- Components of a logistics cycle
- Key logistics terms



INTRODUCTION TO HEALTH COMMODITY SECURITY

What is Health commodity security?

- The process of **ensuring** that everyone has **access** to the quality health commodities they need when they need them
- It is defined over time
- The World Health Organization (WHO) uses the concept of **commodity security** as a starting point for improving the availability of essential health supplies.



INTRODUCTION

Defn:

- Health commodity security **ensures** that **essential** health products—such as medicines, vaccines, diagnostics, and contraceptives—are **available** in the **right quantities**, at the **right time**, at the **right place**, and at **affordable prices**.



INTRODUCTION

It involves:

- Improving the **efficiency, effectiveness**, and **equity** of the supply of health commodities
- Strengthening the government's coordination
- Ensuring the sustainability of the supply of health commodities



WHY HEALTH COMMODITY SECURITY MATTERS:

- **Prevents Stockouts:** Guarantees continuous access to vital health commodities.
- **Improves Health Outcomes:** Ensures that life-saving medications and vaccines are always available for those who need them.
- **Strengthens Health Systems:** Reliable health commodity security supports the broader health infrastructure.



HEALTH COMMODITY SECURITY VS SUPPLY CHAIN MANAGEMENT

- Health commodity security is **the goal** of a health logistics system, which is a supply chain management system that ensures people can access quality health supplies when they need them.

Health commodity security

- The goal of health commodity security is to ensure that people can access quality health supplies when they need them.
- This means that people can choose, obtain, and use the supplies for prevention, treatment, and care.



CONT'D

Supply chain management

- Supply chain management is **the system** that ensures that products are delivered to where they need to go.
- A well-functioning supply chain is essential for health commodity security.
- In addition to supply chain management, other factors that contribute to health commodity security include: **financing, policies, commitment, and supplier base.**



IMPORTANCE OF HEALTH COMMODITY SECURITY

- Commodity security is especially important in low- and middle-income countries (LMICs) where **demand** for healthcare can be **unpredictable**.
- It has a great role in reducing mortality, improving health outcomes, and supporting universal health coverage (UHC).
- Build functional logistics systems, Establish a healthy supplier base, and Consider both internal and external capacities.



TO ENSURE COMMODITY SECURITY, LMICs NEED TO:

- Build **functional logistics systems**, Establish a healthy supplier base, and Consider both internal and external capacities.
- **Category**-specific approaches to commodity security can help identify gaps in access and develop plans to address them.
- For example,
 - Reproductive health commodity security (RHCS)
 - HIV/ AIDS Treatment
 - Anti malaria
 - Anti TB



STRATEGIC PATHWAY TO HEALTH COMMODITY SECURITY

- A strategic pathway refers to the **coordinated, multi-step** approach designed to ensure the **availability** and **reliability** of health commodities across a health system.
- Strategic pathways involve various **stakeholders**, from governments to NGOs to the private sector.
- It focuses on **policy, governance, procurement, logistics, and monitoring**.



IMPORTANCE OF STRATEGIC PATHWAYS

- **Purpose:** Strategic pathways are essential for ensuring **uninterrupted access** to life-saving commodities, **minimizing stockouts**, and **improving health outcomes**.
- **Benefits:**
 - **Prevention of Stockouts:** Reliable supply chains reduce the risk of stockouts.
 - **Cost-Effectiveness:** Better planning leads to optimal use of resources.
 - **Health System Strengthening:** Secure commodities contribute to building stronger health systems.



STAKEHOLDER ROLES IN STRATEGIC PATHWAYS

- **Governments:** Responsible for setting policies, regulating, and funding health commodities.
- **International Organizations:** The World Health Organization (WHO), UNICEF, and the Global Fund provide guidance, funding, and technical assistance.
- **Private Sector:** Pharmaceuticals, logistics providers, and suppliers ensure the production, storage, and distribution of health commodities.
- **NGOs & Civil Society:** Work on the ground to ensure commodities reach underserved populations.



KEY STRATEGIES FOR ENSURING HEALTH COMMODITY SECURITY

○ **Policy Formulation and Governance:**

- Ensure strong policies and governance frameworks to ensure the steady supply of health commodities.
- Regulatory systems to monitor quality and safety.

○ **Procurement Systems:**

- Centralized vs. decentralized models.
- Effective supplier management and competitive bidding processes.

○ **Supply Chain Management:**

- Integrated logistics to move commodities from suppliers to end-users.
- Technology and data-driven solutions (e.g., eLMIS, DAGU, ESW....)

○ **Monitoring and Evaluation:**

- Using key performance indicators (KPIs) to assess the effectiveness of commodity security systems.
- Monitoring for stock levels, wastage, distribution efficiency.



GOVERNANCE AND COORDINATION MECHANISMS

- **Role of Governance:** A strong governance framework is essential to ensure accountability, transparency, and efficiency in commodity management.
- **Coordination Across Sectors:** Health, finance, logistics, and the private sector must be well-coordinated.
- **Multi-Stakeholder Coordination:**
 - National governments lead the process, but it involves collaboration with international partners, NGOs, and private companies.
 - Coordination with local health systems for effective distribution and use.
- **Example:** The **Global Fund** works with governments, NGOs, and the private sector to improve commodity availability for HIV, TB, and malaria.



FINANCIAL STRATEGIES FOR HEALTH COMMODITY SECURITY

- **Financing Health Commodity Security:**
 - Sustainable funding is critical to ensure commodity security.
 - Mechanisms such as government financing, international grants, and private-sector investments play a role.
- **Example of Financial Frameworks:**
 - **Global Health Financing Mechanisms:** GAVI (Global Alliance for Vaccines and Immunization), PEPFAR (U.S. President's Emergency Plan for AIDS Relief).
Example: Program Drugs
 - **Domestic Health Financing:** Increasing national budget allocations for essential medicines.
Example: Revolving drug fund



ROLE OF PROCUREMENT SYSTEMS IN STRATEGIC PATHWAYS

○ **Effective Procurement Strategies:**

- Centralized vs. decentralized procurement: Advantages and challenges of each.
- Forecasting demand to avoid both shortages and overstocking.

○ **Procurement Models:**

- **Centralized:** Government manages procurement centrally to reduce costs.
- **Decentralized:** Local institutions manage their own procurement to ensure faster and more localized decision-making.



SUPPLY CHAIN AND LOGISTICS IN STRATEGIC PATHWAYS

- **Logistics** as a Critical Component:
 - Transport, storage, inventory management, and distribution must be synchronized to ensure timely delivery.
- **Optimizing Supply Chains:**
 - Use of data for **route planning**, **storage optimization**, and **inventory management**.
 - Ensuring that the last mile delivery is effectively managed to reach underserved or remote populations.
- **Example:**
 - Logistics and transport systems used by **UNICEF** to deliver vaccines to hard-to-reach areas.



MONITORING AND EVALUATION SYSTEMS

- **Monitoring Health Commodity Systems:**
 - Continuous monitoring of stock levels, expiration dates, and distribution performance.
 - Use of data to identify bottlenecks and inefficiencies in the system.
- **Key Performance Indicators (KPIs):**
 - Stockouts, lead times, distribution times, and inventory turnover.
 - Regular reporting and auditing to ensure system integrity.
- **Example: LMIS** used by various countries to track health commodities.
 - **DAGU**



CHALLENGES IN THE STRATEGIC PATHWAY

- Weak infrastructure, especially in low-resource settings.
- Inconsistent or unreliable funding.
- Poor coordination among stakeholders.
- Political instability and conflict affecting health commodity flow.



STEPS INVOLVED IN HEALTH COMMODITY SECURITY

- These are steps ensure that the whole system from **policy-making** to **distribution** is well-structured and efficient.
- Health commodity security is achieved through a series of integrated steps, including policy formulation, forecasting, procurement, logistics, monitoring, financial planning, and stakeholder coordination.



STEP 1 – POLICY AND STRATEGIC PLANNING

- **Objective:** Establish clear policies and strategies to guide the entire health commodity security process.
- **Actions:**
 - **National Health Policies:** These policies set the overarching vision for health systems and guide the procurement and distribution of health commodities.
 - **Health Commodity Security Strategy:** Develop strategies at the national and local levels that address specific needs, such as stockouts, quality assurance, and equitable access.
 - **Alignment with Global Goals:** Ensure the strategy aligns with global health frameworks like **Universal Health Coverage (UHC)**.



STEP 2 – FORECASTING AND NEEDS ASSESSMENT

- **Objective:** Accurately predict the required quantities of health commodities to avoid shortages or overstocking.
- **Actions:**
 - **Forecasting Demand:** Use historical data, disease trends, and population demographics to forecast demand.
 - Tools like **LMIS** and **HIS**.
 - **Needs Assessment:** Assess the specific needs of the population based on factors like disease burden, vaccination schedules, and health program priorities (e.g., HIV, malaria).
 - **Data Collection:** Continuously gather data on consumption patterns and changing disease prevalence to refine forecasts.



STEP 3 – PROCUREMENT AND SUPPLIER MANAGEMENT

- **Objective:** Acquire the necessary commodities through an effective and transparent procurement system.
- **Actions:**
 - **Procurement Planning:** Develop a procurement plan based on forecasted needs and available funding.
 - **Supplier Selection:** Use competitive bidding processes to select reliable suppliers who meet quality standards.
 - Criteria: Supplier reliability, price, quality, and delivery capabilities.
 - **Bulk Purchasing:** Consider bulk purchasing agreements to lower costs and ensure timely supply.
 - **Long-term Contracts:** Establish long-term relationships with suppliers for predictable pricing and availability.



STEP 4 – LOGISTICS AND DISTRIBUTION SYSTEMS

- **Objective:** Ensure the effective movement of health commodities from suppliers to health facilities and communities.
- **Actions:**
 - **Warehousing and Storage:** Secure proper storage conditions (e.g., temperature-control) to maintain the quality of health commodities.
 - **Transport and Distribution:** Establish a reliable distribution system that can efficiently move commodities from warehouses to healthcare providers, even in remote areas.
 - **Last-Mile Delivery:** Focus on the last-mile delivery to ensure that commodities reach local clinics and patients, particularly in rural or underserved areas.
 - **Inventory Management:** Implement real-time inventory tracking systems to monitor stock levels and prevent stockouts. LMIS---DAGU



STEP 5 – MONITORING AND EVALUATION

- **Objective:** Track the effectiveness of the health commodity security system and identify areas for improvement.
- **Actions:**
 - **Stock Monitoring:** Regularly monitor stock levels at all points in the supply chain to avoid both shortages and wastage.
 - **Performance Indicators:** Use Key Performance Indicators (KPIs) like on-time delivery rates, fill rates, stockouts, and inventory turnover to assess supply chain performance.
 - **Data Analysis:** Use data analytics to track discrepancies, understand bottlenecks, and make data-driven decisions for continuous improvement.
 - **Regular Audits:** Conduct periodic audits to ensure that there is no diversion of commodities, and the system remains transparent and accountable.



STEP 6 – FINANCIAL SUSTAINABILITY AND RESOURCE ALLOCATION

- **Objective:** Ensure sufficient funding and financial planning for health commodity security.
- **Actions:**
 - **Budget Planning:** Governments and organizations must allocate adequate funds for procurement, logistics, and management of health commodities.
 - **Sustainable Financing:** Secure long-term financing through mechanisms like **government funding**, **international donors**, **public-private partnerships (PPP)**, and **innovative financing models** (e.g., health bonds, health insurance).
 - **Cost-Effectiveness:** Ensure that commodities are procured in the most cost-effective way, without compromising quality.



STEP 7 – STAKEHOLDER COORDINATION AND COLLABORATION

- **Objective:** Facilitate coordination among key stakeholders to ensure smooth operations across the entire health commodity security system.
- **Actions:**
 - **Government Role:** The government is responsible for policy development, regulation, and funding. Ministries of health, finance, and trade should work in tandem to ensure an integrated system.
 - **International Organizations:** Agencies like the **World Health Organization (WHO)**, **UNICEF**, **Global Fund**, and **GAVI** provide technical assistance, funding, and logistical support.
 - **Private Sector:** Pharmaceutical companies and logistics providers are essential for producing, storing, and transporting health commodities.
 - **NGOs and Civil Society:** Work to improve accessibility, especially for marginalized and underserved populations.



STEP 8 – ADDRESSING EMERGING CHALLENGES AND INNOVATIONS

- **Objective:** Adapt the health commodity security system to respond to emerging challenges and incorporate new technologies and approaches.
- **Actions:**
 - **Emerging Diseases:** Adjust procurement and distribution systems to quickly respond to outbreaks (e.g., **COVID-19, Ebola**).
 - **Climate Change:** Consider the impact of climate change on transportation routes, storage facilities, and commodity distribution.
 - **Technology Integration:** Implement innovative technologies such as **RFID tracking, blockchain, artificial intelligence (AI)**, and **mobile health apps** to improve supply chain transparency, data accuracy, and real-time tracking.
 - **Global Partnerships:** Strengthen partnerships between national governments and international organizations to address unforeseen challenges (e.g., pandemics or regional conflicts)



SUMMARY OF STRATEGIC PATHWAY FOR COMMODITY SECURITY

- Pathway 1 – Policy Formulation and Planning
- Pathway 2 – Demand Forecasting and Needs Assessment
- Pathway 3 – Procurement Planning and Supplier Engagement
- Pathway 4 – Logistics and Distribution Systems
- Pathway 5 – Monitoring, Evaluation, and Feedback
- Pathway 6 – Financing and Sustainability
- Pathway 7 – Collaboration, Coordination, and Multi-Stakeholder Engagement
- Pathway 8 – Addressing Emerging Challenges and Innovations



THE LOGISTIC SYSTEM

○ Definition:

- **Logistics System:** A logistics system refers to the process of **planning, implementing,** and **controlling** the efficient flow and storage of health products from the point of origin to the point of use.

○ Role in Health Systems:

- Effective logistics systems ensure that **medicines, vaccines, diagnostics,** and **medical devices** are available in the right quantities, at the right time, and at the right location.



KEY OBJECTIVES OF LOGISTICS IN HEALTH COMMODITY SECURITY

○ Availability of Health Commodities

- Ensures that essential health products are available when and where they are needed, preventing **stockouts** and shortages.
 - Example: Ensuring that vaccines are available for routine immunization campaigns and emergency responses.

○ Cost-Effectiveness

- Manages resources in a way that reduces operational costs, making the system efficient and sustainable.
- Includes the careful management of **inventory, storage, and transportation**.

○ Quality Assurance

- Maintains the integrity and quality of health commodities throughout the logistics chain, including temperature control for sensitive products (e.g., vaccines).
 - Example: Use of **cold chain** systems for maintaining the effectiveness of vaccines.



KEY OBJECTIVE CONT.....

○ **Timeliness of Delivery**

- Ensures timely delivery to health facilities, especially in remote or underserved areas.
 - **Example: Ensuring that life-saving medications for chronic diseases are delivered to clinics without delays.**

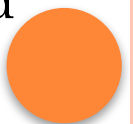
○ **Minimization of Wastage and Loss**

- Prevents the **expiration** and **spoilage** of health commodities through efficient inventory management and proper storage.
- Uses **First-Expire, First-Out (FEFO)** method for managing perishable products.



TECHNOLOGIES SUPPORTING HEALTH LOGISTICS SYSTEMS

- **Automation and Digital Tools:**
 - **Logistics Management Information Systems (LMIS)** help track the flow of commodities and inventory in real-time.
 - **Barcode systems** are used to track inventory and reduce errors.
- **Cold Chain Technology:**
 - Use of **refrigerated transport, solar-powered cold storage, and temperature-monitoring devices** to ensure that vaccines and other temperature-sensitive commodities remain viable.
- **Mobile Health Technologies:**
 - **Mobile apps and SMS tracking systems** for real-time updates and monitoring, improving response times and inventory management.
- **Drones and UAVs:**
 - Drones and Unmanned aerial vehicles (UAVs) are increasingly being used for last-mile deliveries in remote or conflict-affected areas, delivering vaccines, blood, and medical supplies.



BARRIERS TO EFFECTIVE LOGISTICS IN HEALTH COMMODITY SECURITY

Despite the clear purpose of logistics systems, there are several **barriers** that can hinder their effectiveness:

- **Inadequate Infrastructure:**
 - Poor roads, unreliable transportation, and limited access to electricity for cold storage can disrupt the supply chain.
 - **Weak Data Management Systems:**
 - Lack of accurate and real-time data can lead to **overstocking, stockouts, and wastage**.
 - **Financial Constraints:**
 - Insufficient funding for procurement, warehousing, and transportation may limit the ability to maintain a steady supply of health commodities.
 - **Political and Regulatory Barriers:**
 - Delays in regulatory approvals or unstable political environments can disrupt the flow of commodities across borders.
 - **Logistical Challenges in Remote Areas:**
 - Access to remote or conflict-affected regions can be costly and complex, leading to delays in delivery.
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KEY COMPONENTS OF THE LOGISTICS CYCLE

The logistics cycle in health commodity security includes several key components that work together to ensure smooth, efficient operations.

These include:

- 1. Procurement**
- 2. Warehousing and Storage**
- 3. Inventory Management**
- 4. Transportation**
- 5. Distribution**
- 6. Monitoring and Evaluation**



COMPONENT 1 – PROCUREMENT

- **Objective:** The procurement process is the first step in the logistics cycle. It involves acquiring health commodities from suppliers and ensuring the right quantity and quality.
- **Key Actions:**
 - **Supplier Selection:** Identify reliable suppliers based on quality, cost, and delivery capabilities.
 - **Contracting:** Negotiate terms, including prices, delivery schedules, and warranty agreements.
 - **Global and Local Procurement:** Consider both international and local suppliers to balance cost, delivery times, and local availability.



COMPONENT 2 – WAREHOUSING AND STORAGE

- **Objective:** To store health commodities in optimal conditions to prevent damage, contamination, or expiration.
- **Key Actions:**
 - **Temperature-Controlled Storage:** For temperature-sensitive products (e.g., vaccines), use cold chain storage and refrigeration.
 - **Secure Storage:** Use appropriate storage facilities for controlled drugs and valuable commodities (e.g., antibiotics, antiretrovirals).
 - **Stock Management:** Organize commodities according to **First-Expire, First-Out (FEFO)** or **First-In, First-Out (FIFO)** principles to reduce wastage.



COMPONENT 3 – INVENTORY MANAGEMENT

- **Objective:** Maintain accurate and real-time inventory records to ensure that stock levels meet demand without overstocking or stockouts.
- **Key Actions:**
 - **Inventory Tracking:** Use technology such as **RFID tags, barcode scanning, and Electronic Inventory Management Systems (IMS)** for real-time updates.
 - **Replenishment:** Monitor stock levels and reorder products before they run out.
 - **Demand Forecasting:** Use historical data and disease trend forecasts to predict future commodity needs.
 - **Stock Rotation:** Use **FEFO** for perishable goods to reduce the risk of expired products.



COMPONENT 4 – TRANSPORTATION

- **Objective:** The transportation component ensures that health commodities are delivered from the warehouse to health facilities or distribution points in a timely and safe manner.
- **Key Actions:**
 - **Transport Planning:** Plan routes, transportation modes (land, sea, air), and schedules to ensure timely delivery.
 - **Temperature-Controlled Transport:** Ensure that temperature-sensitive items (e.g., vaccines, biologics) are transported in refrigerated vehicles or containers.
 - **Last-Mile Delivery:** Ensure that even remote areas are serviced through effective last-mile delivery strategies (e.g., mobile clinics, drones).



COMPONENT 5 – DISTRIBUTION

- **Objective:** Distribute health commodities to the end-users, which may include health facilities, clinics, or even directly to patients in some cases.
- **Key Actions:**
 - **Health Facility Supply:** Ensure that essential health commodities reach hospitals, clinics, pharmacies, and health posts in a timely manner.
 - **Community-Based Distribution:** In some cases, health commodities may be distributed to community health workers or mobile clinics for outreach.
 - **Emergency Response:** During outbreaks or public health emergencies, distribute commodities rapidly to affected areas.



COMPONENT 6 – MONITORING AND EVALUATION

- **Objective:** Continuously track the effectiveness of the logistics cycle to ensure that health commodities are delivered in the right quantities and on time, while minimizing losses.
 - **Key Actions:**
 - **Stock Monitoring:** Track stock levels at all points in the supply chain, using **Inventory Management Systems (IMS)** or **Logistics Management Information Systems (LMIS)**.
 - **Supply Chain Audits:** Conduct regular audits to ensure the system is functioning as planned and that there is no wastage or diversion of supplies.
 - **Data-Driven Decisions:** Use data analytics to monitor trends and identify bottlenecks in the logistics cycle, such as delays in transportation or warehousing issues.
 - **Feedback Loops:** Create mechanisms to get feedback from users (e.g., healthcare providers) to improve the system.
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INTEGRATION OF COMPONENTS IN THE LOGISTICS CYCLE

- **Flow of Goods:** Each component in the logistics cycle is interconnected. For instance, procurement feeds into warehousing, and inventory management affects transportation scheduling.
- **Technology Integration:** Modern logistics systems use integrated software (e.g., **ERP systems, IMS, LMIS**) to track and manage the entire cycle seamlessly.
- **Coordination:** Effective coordination between procurement, warehousing, transportation, and distribution is crucial for a smooth logistics cycle.



CHALLENGES IN THE LOGISTICS CYCLE

- **Infrastructural Limitations:** Poor roads, lack of transportation infrastructure, and inadequate warehousing facilities can slow down the cycle.
 - **Stockouts and Wastage:** Inefficient inventory management and lack of forecasting can lead to stockouts or excessive wastage of health commodities.
 - **Supply Chain Disruptions:** Natural disasters, political instability, or pandemics can disrupt the logistics cycle, affecting the delivery of essential commodities.
 - **Data Inaccuracies:** Inaccurate or outdated inventory data can lead to mismatched supplies and poor decision-making.
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KEY LOGISTICS TERMS

- **Logistics:** The process of planning, implementing, and controlling the movement and storage of goods, services, and information from the point of origin to the point of consumption to meet the needs of customers or end-users.
- **In Health Commodity Security:** Ensures that health products (vaccines, medicines, medical supplies) are available in the right quantity, at the right time, and in the right condition.
- **Logistics Management:** The management of the flow of goods and services in the supply chain, including storage, transportation, inventory management, and order fulfillment.
- **In Health:** Involves the coordination of various processes to maintain a reliable and efficient supply of essential health commodities.



KEY TERMS

Inventory Management and Stockouts

- **Inventory Management:** The process of overseeing and controlling the storage, movement, and use of inventory within a supply chain to ensure there is enough stock on hand to meet demand while avoiding overstocking.
 - **In Health:** Efficient inventory management ensures essential medicines, vaccines, and other health commodities are always available when needed.
 - **Example:** Use of technologies like **Inventory Management Systems (IMS)** to track stock levels in real-time.
- **Stockouts:** A situation where an item (e.g., medicine or vaccine) is unavailable because the inventory has been depleted.
 - **In Health:** Stockouts of critical health commodities can lead to treatment delays, disease outbreaks, and other public health risks.



KEY TERMS

- **Cold Chain:** A temperature-controlled supply chain required for products that need to be kept at specific temperatures to maintain their effectiveness (e.g., vaccines, biologics).
 - **In Health:** Cold chain logistics ensure that temperature-sensitive commodities (e.g., vaccines) are transported and stored correctly, maintaining their efficacy.
- **First-Expire, First-Out (FEFO):** A system for managing perishable goods based on the expiration dates. The products that are closest to expiration are used or distributed first to avoid wastage.
 - **In Health:** FEFO ensures that health commodities like vaccines and medications are used before they expire, reducing waste.



KEY TERMS

- **Last-Mile Delivery and Lead Time**
- **Last-Mile Delivery:** The final step in the supply chain where health commodities are delivered from a distribution center to the end-user, which may be a healthcare facility, pharmacy, or directly to the patient.
 - **In Health:** Critical for reaching remote areas or underserved communities where access to healthcare may be limited.
- **Lead Time:** The time it takes for an order to be processed and delivered to the end-user. This includes the time taken for procurement, transportation, and delivery.
 - **In Health:** Reducing lead time ensures that critical health commodities are available when needed, especially in emergencies.

